

# Final Homework — Introduction to Data Science

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## Final Homework: Unsupervised Learning, Dimensionality Reduction, and Anomaly Detection

### Motivation

Many problems in data science are fundamentally **ill-defined**. In unsupervised learning, there is often no ground truth telling us how many clusters should exist, which observations should be considered anomalous, or even whether meaningful clusters or anomalies exist at all. Different methods, assumptions, distance measures, and parameter choices may lead to very different conclusions—and more than one conclusion may be reasonable.

The purpose of this exercise is therefore not simply to apply algorithms and report their outputs. You are expected to investigate the data, question your assumptions, compare competing methods, and critically evaluate disagreements between them. Most importantly, you should connect the mathematical results to the real-world meaning of the data.

In an ill-defined problem, finding an answer is only part of the task. The greater challenge is understanding **why you trust that answer, when you should question it, and what its limitations are**.

### Objective

The goal of this assignment is to explore the structure of real-world data using unsupervised learning techniques.

Unlike supervised learning, no labels guide the analysis. Your task is to investigate the geometry, hidden structure, clusters, and abnormal behaviors within the dataset.

This assignment combines:

- Clustering methods
- Dimensionality reduction
- Principal Component Analysis (PCA)
- Multi-dimensional anomaly detection
- Critical interpretation of results

You are expected not only to apply algorithms, but also to explain:

- What the algorithms discovered
- Why they behave differently

- What assumptions they make
- What limitations they have

## 1 Dataset Selection (10%)

### Requirements

Choose a dataset satisfying all of the following:

- At least 1000 rows
- At least 8 features
- Mostly numerical variables
- Real-world dataset
- No pre-labeled anomaly column

You may use datasets from:

- Kaggle
- UCI Repository
- OpenML
- Government/open-data repositories
- Cybersecurity logs
- Sensor data
- Financial data
- Medical measurements
- IoT systems

Students may either use a dataset from one of the previous homework assignments or choose a new dataset. The dataset should be selected in accordance with the project requirements.

### You Must Describe

- Where the dataset comes from
- Why it was collected
- Who collected it (if known)
- What each feature represents
- Possible limitations or biases in the data

## 2 Exploratory Data Analysis (15%)

Perform a complete exploratory analysis.

## Structural Analysis

- Number of rows and columns
- Missing values
- Data types
- Statistical summary
- Correlation matrix

## Distribution Analysis

For each major feature, include:

- Histogram
- Boxplot
- Skewness
- Kurtosis

## Outlier Exploration

Discuss:

- Heavy tails
- Extreme values
- Possible noise
- Measurement artifacts
- Data-entry problems

## 3 Dimensionality Reduction (20%)

### 3.1 Principal Component Analysis (PCA)

Apply PCA to the dataset.

You must include:

- Explained variance ratio
- Cumulative explained variance
- Scree plot
- Projection into:
  - 2D
  - 3D

Discuss:

- How many principal components are needed to adequately represent the data?

- How much information is lost when reducing the dimensionality?
- Which variables are strongly correlated, and how does this affect the PCA results?
- Interpret the major principal components. Which original variables contribute most to each component?
- Does the reduced dataset provide a more effective representation of the data than the original dataset? In your answer, discuss the level of redundancy in the original data.

Discuss:

- Local vs global structure
- Cluster separation
- Stability
- Computational cost
- Interpretability
- Redundancy

## 4 Clustering Analysis (20%)

Apply at least three clustering methods.

Possible methods:

- K-Means
- DBSCAN
- Hierarchical Clustering

### Clustering on the Feature Space

Apply clustering not only to the samples, but also to the feature space itself (that is, to the transpose of the data matrix).

Discuss the following questions:

- What information can be obtained by clustering the features?
- Did the analysis reveal groups of related or redundant features?
- Were there features that appeared isolated or significantly different from the others?
- Did this analysis provide insights that were not visible when clustering the samples directly?
- In what types of problems or domains might clustering features be particularly useful?

Explain whether clustering the feature space helped uncover hidden structure, dependencies, correlations, or potential feature hierarchies within the dataset.

### Required Analysis

For each method:

## **Explain**

- Explain the mathematical intuition behind the clustering method.
- Discuss the main assumptions of the method.
- Describe the main strengths and weaknesses of the method.
- Do the resulting clusters have a meaningful real-world interpretation? Explain what each cluster may represent in the context of the dataset.

## **Evaluate**

Use clustering metrics, the average variance in each cluster vs the global variance. Is this metric suitable? If so, explain. Otherwise, provide an alternative.

## **Visualize**

Show clusters on PCA projection.

## **Discussion**

Discuss:

- Why different algorithms produce different clusters
- Sensitivity to hyperparameters
- Cluster geometry
- Density-based vs centroid-based approaches
- Situations where clustering may fail

# **5 Multi-Dimensional Anomaly Detection (25%)**

This is the main part of the assignment.

You must apply and compare the following anomaly detection methods.

## **5.1 Z-Score Method**

Apply a classical Z-score method.

Discuss:

- Assumption of Gaussianity
- Why this approach may fail in high dimensions
- Feature-wise vs global anomaly detection

## 5.2 Isolation Forest

Apply Isolation Forest.

Analyze:

- Explain the isolation mechanism used by the algorithm.
- Explain the role of random partitioning in identifying anomalous observations.
- Discuss the contamination parameter and its effect on the results.
- Discuss the advantages and limitations of the method, particularly in high-dimensional datasets.
- Explain how the anomaly score is calculated and interpret what a high or low score represents.

Visualize detected anomalies.

## 5.3 Local Outlier Factor (LOF)

Apply LOF.

Discuss:

- Local density estimation
- Neighborhood effects
- Sensitivity to parameter  $k$
- Difference between global and local anomalies

Visualize anomalies.

## 6 Comparison Section

Compare all anomaly detection methods.

The discussion should address:

- Which anomalies are identified by multiple methods? Discuss the level of agreement between the methods.
- Which anomalies are method-specific? Explain why different methods may identify different observations as anomalous.
- Compare the sensitivity of the methods to feature scaling.
- Compare the sensitivity of the methods to the dimensionality of the data.
- Compare the computational runtime and scalability of the methods.
- Discuss the robustness of each method to noise in the data.
- Compare the interpretability of the methods and their results.
- Discuss false positives and false negatives in the context of your dataset. Explain what each type of error means in real-world terms and what its potential consequences are. Your explanation should be understandable to a non-technical audience.

## 7 Critical Analysis and Reflection (10%)

Write a discussion section addressing the following topics:

- Explain why anomaly detection is fundamentally a difficult and ill-defined problem.
- Discuss the curse of dimensionality and its effect on anomaly detection methods.
- Explain why distance-based measures may become less meaningful as the dimensionality of the data increases.
- Discuss the difference between noise and a genuine anomaly. Provide an example from your dataset of an observation that may be detected as anomalous but is nevertheless a legitimate instance.
- Discuss the risks of assuming that the data follow a Gaussian distribution. What types of anomalies might be incorrectly identified or missed because of this assumption?
- Explain why visualizations obtained after dimensionality reduction (e.g., PCA or any other dimension reduction technique) may provide a misleading picture of anomalies or clusters in the original high-dimensional space.

### Ethical Considerations

Discuss ethical considerations such as:

- Discuss the real-world consequences of false alarms. When might too many false alarms make an anomaly detection system ineffective?
- Discuss the use of anomaly detection in surveillance. What are the potential benefits and risks?
- Discuss anomaly detection in medical applications. What could be the consequences of incorrectly classifying a patient as anomalous or normal?
- Discuss anomaly detection in cybersecurity. Compare the potential consequences of missing a genuine attack with those of incorrectly flagging legitimate activity as malicious.
- Discuss the ethical considerations associated with anomaly detection. In particular, consider privacy, bias, fairness, and the consequences of labeling individuals or behaviors as “abnormal.”

## 8 Final Visualization Dashboard (Optional Bonus +5%)

Build a small dashboard or notebook section containing:

- PCA visualization
- Cluster visualization
- Interactive anomaly exploration
- Pairplots
- Heatmaps

## 9 Submission Requirements

Submit the following:

### Report

The report must include:

- Clearly describe the methodology used in your analysis and justify your main methodological choices.
- Provide the mathematical intuition and relevant mathematical explanations behind the methods used.
- Use appropriate visualizations to support your analysis. Each visualization should have a clear purpose and be properly explained.
- Interpret the results in the context of the dataset and the real-world problem. Do not limit the discussion to numerical results.
- Provide a critical discussion of your results, including limitations, assumptions, unexpected findings, and possible alternative explanations.

This section should be presented inside text boxes within the notebook

### Code

Provide:

- Clean notebook or scripts
- Reproducible pipeline
- Comments and explanations
- Meaningful functions and variables names

## Important Notes

You are **NOT** graded on obtaining “beautiful” clusters.

You are graded on:

- Demonstrate a deep understanding of the methods used, including their mathematical intuition, assumptions, and limitations.
- Demonstrate critical thinking and analytical reasoning rather than simply applying algorithms and reporting their outputs.
- Interpret the results in the context of the dataset and explain their practical significance.
- Apply the methods with appropriate methodological rigor, including preprocessing, parameter selection, validation, and comparison between methods.
- Communicate the analysis clearly and effectively. The main conclusions should be understandable to both technical and non-technical audiences.

In real-world data:



- Consider the possibility that meaningful clusters may not exist in the dataset. Do not assume that every dataset naturally contains a clear clustering structure.
- Discuss cases in which the distinction between normal and anomalous observations is ambiguous or depends on the context.
- Different methods may produce different results. Discuss possible reasons for disagreement between methods and explain which results you consider more meaningful for your dataset.

Understanding *why* this happens is an essential part of the assignment.

It is important to note that the inability to identify meaningful clusters, detect significant anomalies, or achieve effective dimensionality reduction does not necessarily indicate poor analysis. In some cases, the underlying structure may simply not exist within the dataset. If this is the case, you should clearly explain your findings and provide a well-reasoned justification based on the characteristics of the data and the methods applied.

## Submission Deadline

**18.9.2026**

## Good Luck

**Data science is not only about building models.**

It is about learning how to reason under uncertainty, work with incomplete information, question imperfect assumptions, and critically evaluate the conclusions we draw from data.

**Good luck, and enjoy the exercise!**